

# Software Requirements Specification (SRS)

## AI-Powered Multi-Source Data Intelligence & Conversational Analytics Platform

**Version:** 1.0

**Document Type:** Software Requirements Specification (SRS)

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### 1. Introduction

#### 1.1 Purpose

The purpose of this application is to provide an enterprise AI-powered data intelligence platform capable of connecting to multiple heterogeneous data sources simultaneously, allowing users to perform conversational AI-based analysis using natural language (text and voice), generate dashboards, reports, documents, and insights without requiring SQL or technical knowledge.

The platform acts as a semantic AI layer over structured and unstructured enterprise data.

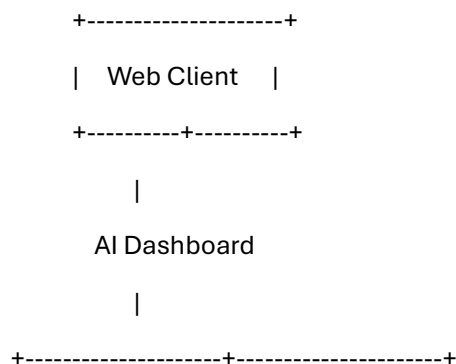
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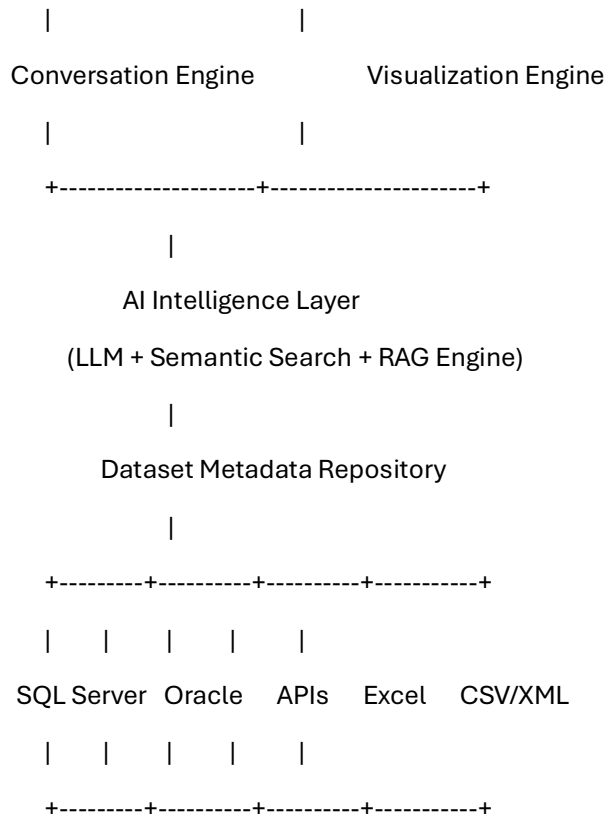
#### 1.2 Objectives

The system shall:

- Connect simultaneously to multiple databases and external systems.
  - Combine data from different sources for unified AI analysis.
  - Allow users to ask business questions in natural language.
  - Understand which datasets/tables are relevant to the question.
  - Maintain conversational memory.
  - Generate dashboards, charts, documents and reports.
  - Provide enterprise-grade security and permissions.
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### 2. High Level Architecture





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### 3. Functional Modules

| Module                   | Description                                 |
|--------------------------|---------------------------------------------|
| User Management          | Users, Roles, Authentication                |
| Data Source Manager      | Create and manage multiple data connections |
| Dataset Manager          | Temporary datasets and metadata             |
| AI Engine                | Conversational AI                           |
| Semantic Metadata Engine | Maps questions to datasets                  |
| Dashboard                | Visual analytics                            |
| Report Generator         | PDF, DOCX, Excel                            |
| Voice Interface          | Speech-to-Text and Text-to-Speech           |
| Security                 | Permissions                                 |
| Audit Logs               | Activity monitoring                         |

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## **4. Functional Requirements**

### **Module 1 – User & Security Management**

#### **Features**

##### **FR-1 User Registration**

System shall allow creation of users.

Fields

- Username
  - Full Name
  - Email
  - Mobile
  - Password
  - Status
  - Default Language
  - Time Zone
- 

##### **FR-2 Email Verification**

Upon user creation:

- Verification Email
  - Email OTP
  - Password Reset Link
- 

##### **FR-3 Password Management**

- Forgot Password
  - Reset Password
  - Password Expiry
  - Password History
  - MFA (Optional)
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##### **FR-4 Roles**

One user can belong to multiple roles.

Example

Finance

Sales

HR

Operations

CEO

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#### **FR-5 Category Permissions**

Permissions are assigned on Categories rather than database connections.

Example

##### **Category Allowed**

Finance Yes

Sales No

Inventory Yes

The AI shall only search datasets within authorized categories.

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#### **Module 2 – Data Source Management**

##### **Supported Data Sources**

The application shall support:

- Microsoft SQL Server
- Oracle
- MySQL
- PostgreSQL
- MariaDB
- SQLite
- Excel
- CSV
- XML
- JSON
- REST APIs
- SOAP APIs

- ODBC
- JDBC
- SharePoint Files
- OneDrive
- Google Sheets (Future)
- Azure Storage
- Amazon S3

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### **Connection Definition**

Each connection shall contain:

Connection Name

Source Type

Server

Database

Authentication Type

Username

Password

Encrypted Credential

Category

Module

Refresh Frequency

Status

Description

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### **Authentication Methods**

- SQL Authentication
- Windows Authentication
- OAuth2
- API Key
- JWT Token
- Basic Authentication

- Bearer Token

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### **Multiple Concurrent Connections**

System shall allow unlimited concurrent connections.

Example

Finance SQL

Sales Oracle

CRM API

HR Excel

Projects API

Inventory SQL

All active simultaneously.

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### **Module 3 – Dataset Manager**

Each connection may create one or more temporary datasets.

Example

Finance\_2026

Sales\_Current

Inventory\_HYD

Projects\_Q2

HR\_Payroll

Each dataset shall contain:

Dataset ID

Dataset Name

Category

Module

Source

Refresh Date

Owner

Record Count

Schema

Temporary Cache Location

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### **Dataset Sources**

Dataset may originate from

- Table
  - View
  - SQL Query
  - API
  - Excel Sheet
  - CSV
  - Multiple Tables
- 

### **Module 4 – Category & Module Definition**

Each connection must define

Category

Example

Finance

HR

Sales

Inventory

Projects

Manufacturing

CRM

Procurement

Module

Example

Accounts Receivable

General Ledger

Payroll

Sales Orders

Inventory

Purchase Orders

The AI shall utilize Category + Module metadata to determine relevant datasets.

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### **Module 5 – Data Selection**

For every connection, users shall be able to select:

Specific Tables

Specific Views

Custom SQL Query

Stored Procedure

API Endpoint

Excel Sheet

CSV File

The selected data shall become an AI Dataset.

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### **Module 6 – AI Semantic Engine**

This is the core module.

The AI shall automatically identify

- Relevant Category
- Relevant Module
- Relevant Dataset
- Relevant Tables
- Relevant Columns
- Relationships

without requiring user knowledge.

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Example

User asks

Which customers have outstanding invoices above 5 Lakhs?

AI determines

Category

Finance

Module

Accounts Receivable

Dataset

Finance\_2026

Tables

Customer

Invoice

Receipt

Outstanding

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Another question

Compare these customers with Sales Orders.

AI switches to

Sales Dataset

joins Customer Master

returns comparison.

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## **Module 7 – Multi Dataset Analysis**

AI shall combine datasets.

Example

Finance SQL

- 

CRM API

- 

Excel

Example Query

Which customers have overdue invoices and unresolved CRM complaints?

The AI shall automatically merge datasets.

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Example

Inventory SQL

- 

Vendor Excel

- 

Purchase API

Question

Which vendors delayed delivery and currently have low inventory?

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## **Module 8 – Conversational AI**

Continuous conversation.

Example

User

Show unpaid invoices.

AI

Would you like all branches or Hyderabad only?

User

Only Hyderabad.

AI

Should I include overdue only?

User

Yes.

AI

Displays report.

Conversation context remains active.

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Supported Input

Text

Voice

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## Supported Output

Text

Tables

Charts

Graphs

Voice

PDF

Word

Excel

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## Module 9 – AI Suggested Questions

AI shall proactively ask follow-up questions.

Example

User

Show sales.

AI

Would you like

Monthly

Quarterly

Yearly

Region-wise

Product-wise

Salesperson-wise

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## Module 10 – AI Dashboard

Dashboard components

Cards

KPIs

Tables

Pivot

Charts

Heat Maps

Geo Maps

Tree Maps

Pie

Bar

Line

Area

Gauge

Scatter

Forecast

Drill Down

Drill Through

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Dashboard Output

Interactive

Exportable

Printable

Shareable

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## **Module 11 – Reporting Engine**

Supported formats

PDF

DOCX

Excel

CSV

HTML

JSON

XML

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Reports may contain

Narrative Summary

Charts

AI Recommendations

Tables

Appendix

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## **Module 12 – Export & Communication**

Users shall export

PDF

DOCX

Excel

CSV

Image

PowerPoint (Future)

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Users shall

Print

Email

Download

Schedule Reports

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Email Options

Recipient

Subject

Body

Attachment

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## **Module 13 – Voice Interface**

Voice Input

Speech Recognition

Natural Language

Multiple Languages

Noise Filtering

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Voice Output

AI Response

Read Reports

Read Summary

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#### **Module 14 – Metadata Engine**

Maintains

Schema

Relationships

Keys

Descriptions

Business Glossary

Categories

Modules

Refresh History

This enables semantic AI understanding.

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#### **Module 15 – Audit Trail**

Capture

Login

Logout

Dataset Creation

Queries

AI Conversations

Exports

Print

Email

Errors

## **5. Database Design**

### **Master Tables**

Users

Roles

RolePermissions

Categories

Modules

Connections

ConnectionAuthentication

Datasets

DatasetTables

DatasetColumns

BusinessGlossary

ConversationHistory

AIFeedback

AuditLogs

RefreshSchedules

ReportTemplates

DashboardTemplates

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## **6. AI Functional Requirements**

The AI shall:

- ✓ Understand natural language
- ✓ Understand business terminology
- ✓ Automatically identify relevant datasets
- ✓ Combine multiple sources
- ✓ Generate SQL dynamically
- ✓ Generate API requests

- ✓ Analyze Excel files
  - ✓ Analyze uploaded documents
  - ✓ Maintain conversation context
  - ✓ Ask clarification questions
  - ✓ Recommend further analysis
  - ✓ Explain calculations
  - ✓ Summarize findings
  - ✓ Detect anomalies
  - ✓ Predict trends (future enhancement)
  - ✓ Generate business recommendations
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## **7. Security Requirements**

### **Authentication**

- Username/Password
  - Email Verification
  - MFA
  - OAuth
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### **Authorization**

Permissions based upon

Role

Category

Module

Dataset

Dashboard

Reports

Export

Administration

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### **Encryption**

TLS 1.3

AES-256 Credentials

Encrypted Password

Encrypted Tokens

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**Audit**

Every AI question

Every Export

Every Print

Every Email

Every Login

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**8. Non-Functional Requirements**

| Requirement               | Target                                         |
|---------------------------|------------------------------------------------|
| Concurrent Users          | 10,000+                                        |
| Simultaneous Data Sources | Unlimited (subject to infrastructure)          |
| Supported Dataset Size    | 100+ million records per dataset               |
| Average AI Response Time  | <5 seconds for indexed datasets                |
| Dashboard Load Time       | <3 seconds                                     |
| Availability              | 99.9%                                          |
| Scalability               | Horizontal scaling with distributed processing |
| Backup                    | Daily incremental, weekly full                 |
| Localization              | Multi-language support                         |
| Accessibility             | WCAG 2.1 AA compliant                          |

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**9. Future Enhancements**

- AI-generated dashboards from plain-language prompts.
- Automated schema discovery and relationship inference.
- Scheduled AI monitoring with anomaly alerts.

- Predictive analytics and forecasting models.
  - Integration with collaboration platforms such as Microsoft Teams and Slack.
  - AI agents capable of executing approved business workflows.
  - Mobile application with voice-first interaction.
  - Data lineage visualization and impact analysis.
  - Semantic business glossary with user-defined terminology.
  - Plugin framework for custom connectors and domain-specific AI skills.
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## **10. Success Criteria**

The solution will be considered successful when it can:

- Connect to multiple heterogeneous data sources simultaneously with secure authentication.
- Organize datasets using business categories and modules for semantic discovery.
- Automatically identify relevant datasets, tables, and relationships based on natural language questions.
- Analyze structured data (databases) and unstructured/semi-structured files together within a single AI conversation.
- Maintain contextual, multi-turn conversations using both text and voice.
- Generate responses as narrative text, interactive data grids, visual dashboards, charts, and downloadable PDF/DOCX reports.
- Enforce role-based access so users can only query authorized categories and datasets.
- Allow secure export, printing, and emailing of AI-generated outputs with complete audit logging.

This specification provides a strong foundation for an enterprise-grade AI Data Intelligence Platform that can evolve into a semantic business intelligence solution, enabling users to interact with organizational data through natural conversation rather than traditional reporting tools.